

Linux Basics and HPCC Cluster Intro



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Agenda

- Cluster Introduction
- Accessing the Cluster
- Storage
- Linux commands
- Module System
- Environment Variables
- Submitting Cluster Jobs
- Scripting
- Monitoring the Cluster Jobs
- Helpful Resources

High Performance Computing Center

- Purpose
 - Provide High Performance Computing and Bigdata Resources to support and enable research at UCR.
- Access
 - Subscription based membership.
 - Accessible both on and off campus.
 - Organized by lab and department affiliation.

HPCC Website

Website Address: hpcc.ucr.edu



- Home ▲
- Introduction
- Activity report
- Slides
- News ▼
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- Documents ▼
- Contacts ▼
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- Hardware ▼
- Software ▼
- Manuals ▼

High-Performance Computing Center (HPCC)

Summary: welcome to the website of the High-Performance Computing Center (HPCC) at UC Riverside. This site gives an overview of the HPC resources and services provided by our center.

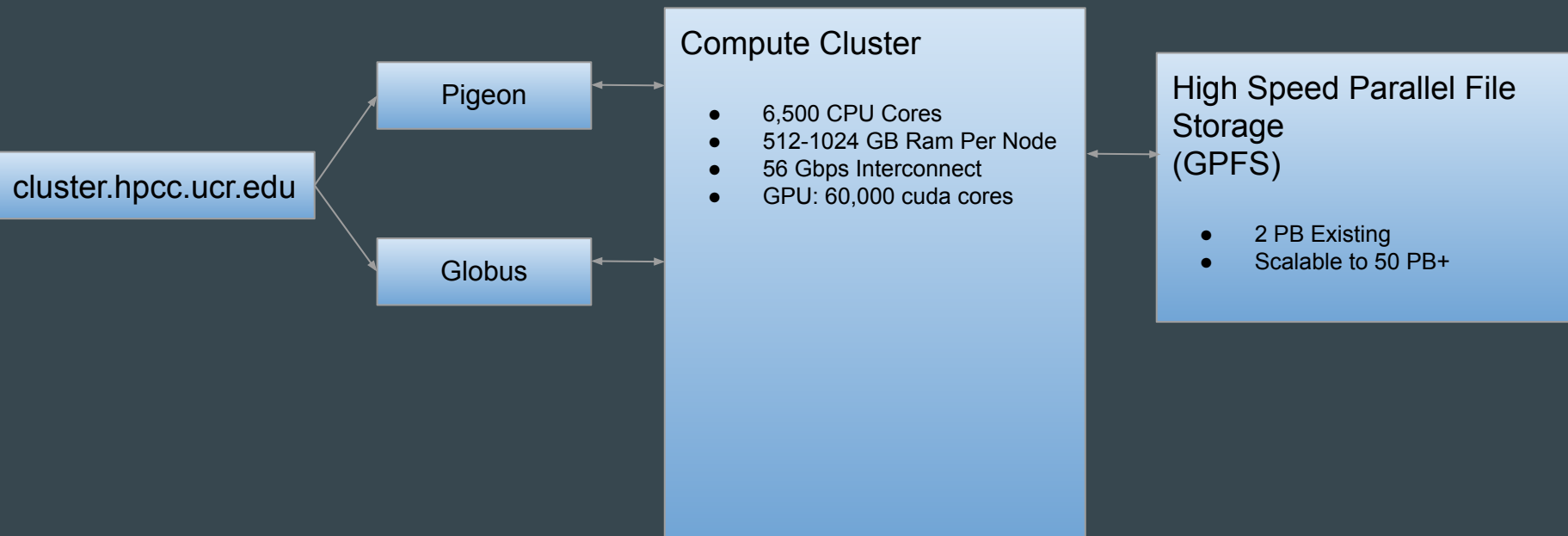
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Mission and Services

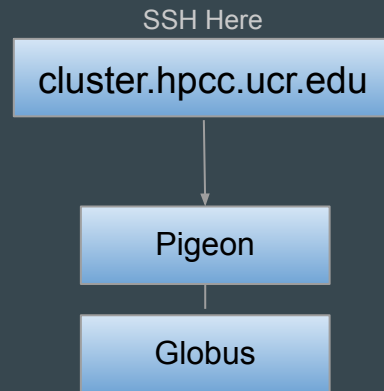
The High-Performance Computing Center (HPCC) provides state-of-the-art research computing infrastructure and training accessible to all UCR researchers and affiliates at low cost. This includes access to the shared HPC resources and services summarized below. The main advantage of a shared research computing environment is access to a much larger HPC infrastructure (with thousands of CPUs/GPUs and many PBs of directly attached storage) than what smaller clusters of individual research groups could afford, while also providing a long-term sustainability plan and professional systems administrative support.

HPCC Cluster Overview

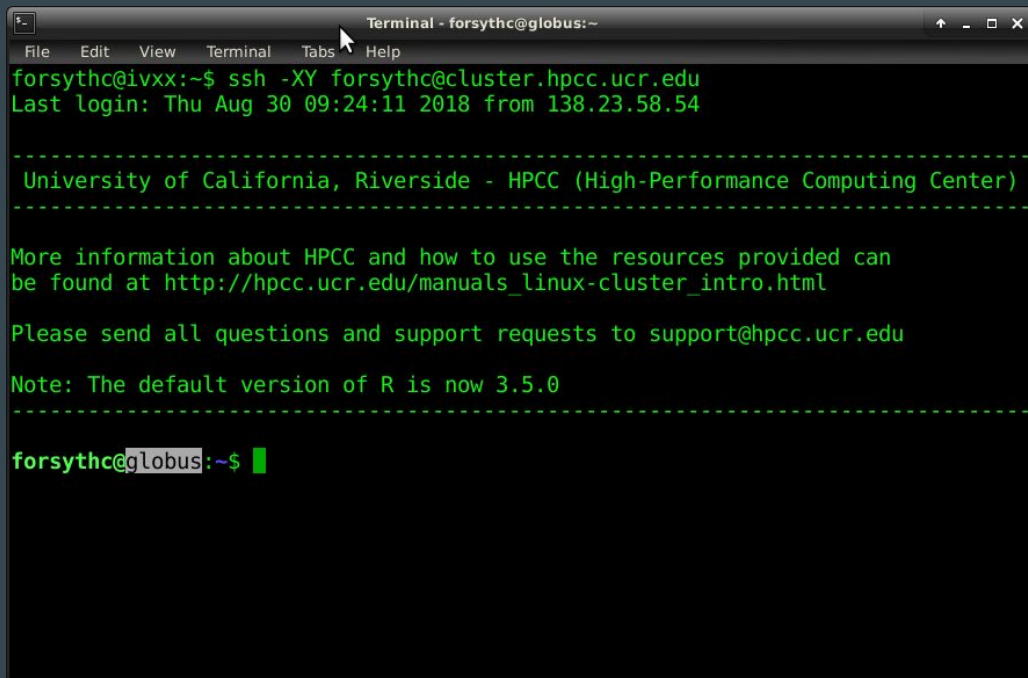


Accessing the Cluster - SSH

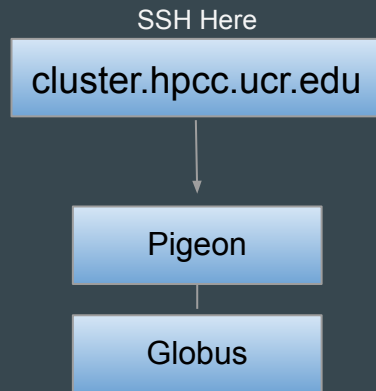
- SSH is the main protocol used to interface with the HPCC Cluster.
- Start by ssh'ing from your system to `cluster.hpcc.ucr.edu`
 - Windows
 - Putty
 - `cluster.hpcc.ucr.edu`
 - MobaXterm
 - `ssh -XY username@cluster.hpcc.ucr.edu`
 - Mac
 - Terminal
 - `ssh username@cluster.hpcc.ucr.edu`
 - Linux
 - Terminal
 - `ssh -XY username@cluster.hpcc.ucr.edu`
 - The “-XY” allows graphical output to be passed back to your system.
- `cluster.hpcc.ucr.edu` is a special DNS name that load balances incoming connections to one of the available login nodes.



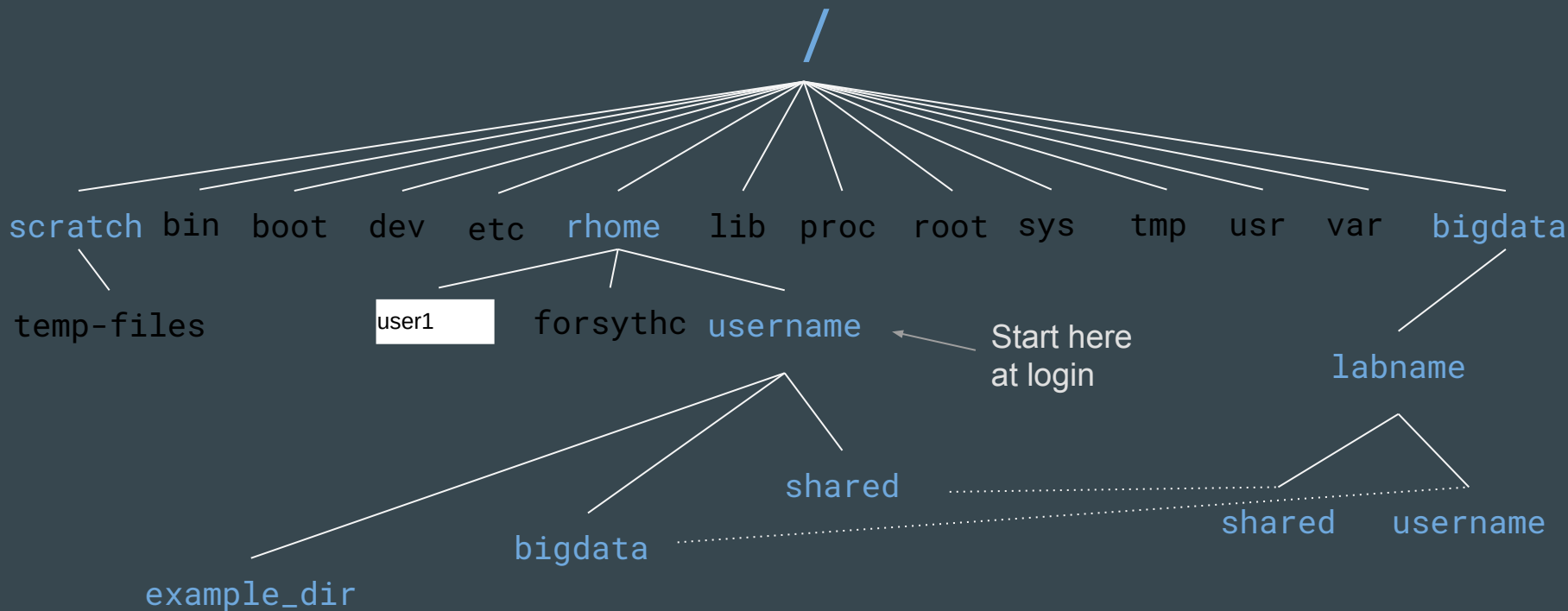
Accessing the Cluster - SSH



```
Terminal - forsythc@globus:~  
File Edit View Terminal Tabs Help  
forsythc@ivxx:~$ ssh -XY forsythc@cluster.hpcc.ucr.edu  
Last login: Thu Aug 30 09:24:11 2018 from 138.23.58.54  
  
-----  
University of California, Riverside - HPCC (High-Performance Computing Center)  
-----  
  
More information about HPCC and how to use the resources provided can  
be found at http://hpcc.ucr.edu/manuals\_linux-cluster\_intro.html  
  
Please send all questions and support requests to support@hpcc.ucr.edu  
  
Note: The default version of R is now 3.5.0  
-----  
forsythc@globus:~$
```



Storage - Directory Structure



Storage - Directory Structure

Home is used for scripting, debugging and small files. (20 GB Quota)

`/rhome/username`

Bigdata is used for parallel jobs, high read/write operations and big files. (100 GB for 100/yr, 10 TB for 1000/yr)

`/bigdata/labname/username`

`/bigdata/labname/shared`

Note: All lab members share the same bigdata pool.

Scratch is local to the compute node and temporary (30 days max).

`/scratch`

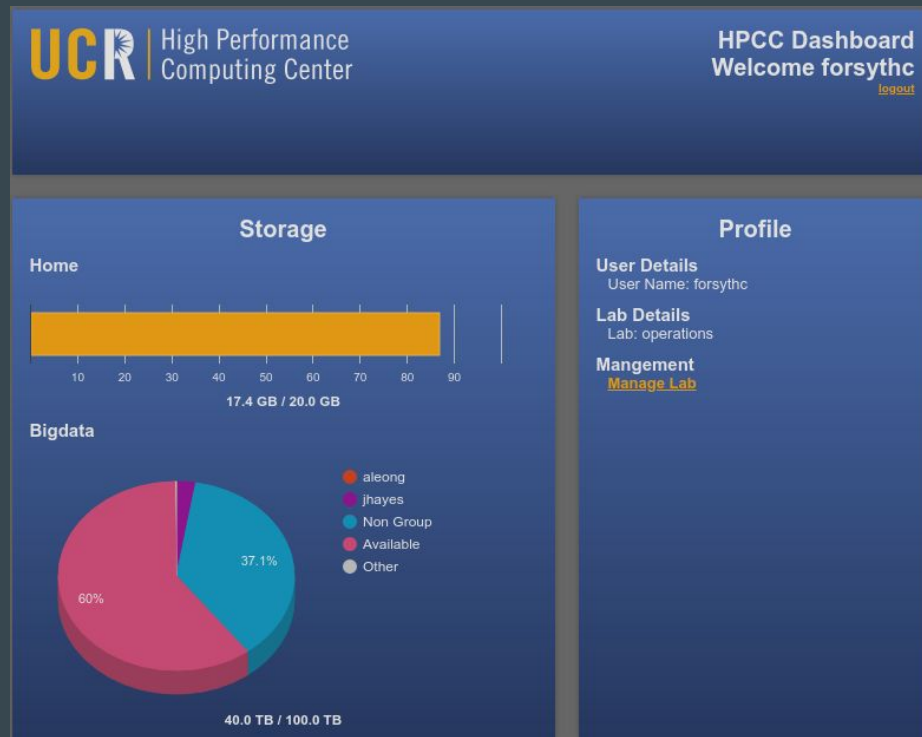
Storage - Directory Structure

```
Terminal - tmux.2.ssh
File Edit View Terminal Tabs Help
repo@penguin:~$ ls
bigdata shared
repo@penguin:~$ ll
total 0
lrwxrwxrwx 1 repo operations 24 Nov 20 2017 bigdata -> /bigdata/operations/repo
lrwxrwxrwx 1 repo operations 26 Nov 20 2017 shared -> /bigdata/operations/shared
repo@penguin:~$ ls -latr
total 768
-rw-r--r-- 1 repo operations 677 Apr 8 2013 .profile
-rw-r--r-- 1 repo operations 92 Jul 11 2015 .bashrc
-rw-r--r-- 1 repo operations 269 Oct 31 2016 .Rprofile
-rw-r--r-- 1 repo operations 12104 Oct 31 2016 .vimrc
-rw-r--r-- 1 repo operations 4882 Oct 31 2016 .tmux.conf
drwxr-xr-x 18 repo operations 4096 Nov 1 2016 .vim
lrwxrwxrwx 1 repo operations 24 Nov 20 2017 bigdata -> /bigdata/operations/repo
lrwxrwxrwx 1 repo operations 26 Nov 20 2017 shared -> /bigdata/operations/shared
-rw----- 1 repo operations 7785 Nov 30 2017 .viminfo
drwx----- 2 repo operations 4096 Jan 11 2018 .ssh
drwx----- 5 repo operations 4096 May 4 16:52 .
drwxr----- 3 repo operations 4096 May 4 16:52 .pki
-rw----- 1 repo operations 9149 Jul 18 14:21 .bash_history
drwxr-xr-x 1072 root root 262144 Sep 5 19:18 ..
repo@penguin:~$ █
[ aegis ] 1:ssh- 2:ssh* [ 0.00 0.01 0.05 ][ Thu 2018-09-06 09:22 ]
```

Storage - Directory Structure

How much data am I using currently:

- <https://dashboard.hpcc.ucr.edu>
- check_quota home
- check_quota bigdata



Storage - Uploading/Downloading

- Moving files to or from the Cluster is allowed via SCP or SFTP.
- SCP
 - `scp <from> <to>`
 - `scp user@remote_host:filename .`
 - Copies file from server to local machine (typed from local machine prompt). The “.” copies to pwd, you can specify here any directory, use wildcards to copy many files.
 - `scp filename user@remote_host:~/dir/.`
 - Copies file from local machine to server.
 - `scp -r user@remote_host:directory/ ~/dir`
 - Copies entire directory from server to local machine.
- SFTP
 - FileZilla is a graphical SFTP client available free for Linux, Mac and Windows platforms.
 - <https://filezilla-project.org/>
 - The SFTP protocol is also available from the command line terminal in Linux and Mac.

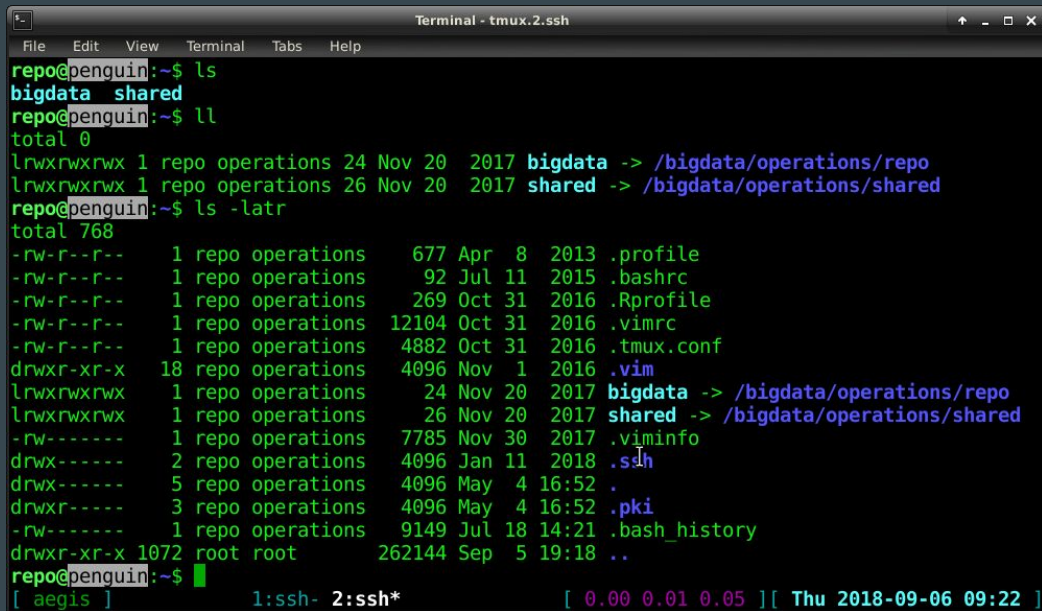
Storage - Filesystem

- Case sensitive
 - All paths and commands are case sensitive, an uppercase letter is not the same as a lowercase letter.
- Path Types
 - Absolute path
 - Full path from top to bottom
 - `/rhome/yourname/example_dir/example_file`
 - Relative path
 - Current working directory is implied
 - `./example_dir/example_file`
 - `../another_dir/example_dir/example_file`

Linux - File Permissions

To view the permissions on the files in your current directory:

- List files
 - `ls`
- List files in long format
 - `ls -l`
 - `ll`
- List current directory in long format
 - `ls -ld`
- List all files in long format
 - `ls -la`



A terminal window titled "Terminal - tmux.2.ssh" showing a series of commands and their outputs. The user is in a directory where they can see two subdirectories: "bigdata" and "shared". The terminal shows the output of `ls`, `ll`, and `ls -latr`. The `ls -latr` output shows a detailed list of files and directories with their permissions, sizes, dates, and names. The files include `.profile`, `.bashrc`, `.Rprofile`, `.vimrc`, `.tmux.conf`, `.vim`, `.viminfo`, `.ssh`, `.pki`, `.bash_history`, and `..`. The terminal also shows the user's prompt and the current directory path.

```
repo@penguin:~$ ls
bigdata shared
repo@penguin:~$ ll
total 0
lrwxrwxrwx 1 repo operations 24 Nov 20 2017 bigdata -> /bigdata/operations/repo
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-rw----- 1 repo operations 9149 Jul 18 14:21 .bash_history
drwxr-xr-x 1072 root root 262144 Sep 5 19:18 ..
repo@penguin:~$
```

Linux - File Permissions

drwxrwxrwx

Directory User Group Other

Every file and directory has permissions:

- User read/write/execute
- Group read/write/execute
- Other read/write/execute

Change the permissions of a file:

- `chmod g+w filename`
- `chmod o-r filename`
- `chmod u+x filename`

```
Terminal - tmux.2.ssh
File Edit View Terminal Tabs Help
repo@penguin:~$ ls
bigdata shared
repo@penguin:~$ ll
total 0
lrwxrwxrwx 1 repo operations 24 Nov 20 2017 bigdata -> /bigdata/operations/repo
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-rw----- 1 repo operations 9149 Jul 18 14:21 .bash_history
drwxr-xr-x 1072 root root 262144 Sep 5 19:18 ..
repo@penguin:~$
```

[aegis] 1:ssh- 2:ssh* [0.00 0.01 0.05] [Thu 2018-09-06 09:22]

Linux - Basic Commands

Basic Commands

- `pwd` - Print working directory
- `ls` - List files in directory
- `touch` - Make an empty file
- `mkdir` - Make a directory
- `cd` - Change to directory
- `cp` - Copy file[s] from a directory to a directory
- `mv` - Move file[s] from a directory to a directory
- `rm` - Remove a file
- `rmdir` - Remove an empty directory

Note: `ctrl+c` will cancel a running command

Screen / Tmux (don't forget to 'exit' or 'kill' your sessions)

Screen - quick reference

- Open a new session
`screen`
- Detach from a session
`C-a d`
- Resume a previous session
`screen -rd`
- Open new window
`C-a c`
- Rename window
`C-a A`
- Navigate to previous or next window
`C-a p` OR `C-a n`
- Window list
`C-a "`
- Split window horizontally or vertically
`C-a S` OR `C-a |`
- Navigate window panes
`C-a tab`
- Close window pane
`C-a X`

Tmux - quick reference (screen/vi mode)

- Open a new session
`tmux`
- Detach from a session
`C-a d`
- Resume a previous session
`tmux attach -d`
- Open new window
`C-a c`
- Rename window
`C-a ,`
- Navigate to previous or next window
`C-a p` OR `C-a n`
- Window list
`C-a w`
- Split window horizontally or vertically
`C-a "` OR `C-a %`
- Navigate window panes
`C-a arrow-key`
- Close window pane
`C-a s`

Linux - Text editors and viewers

Viewers

- `less`
- `more`

Editors

- `nano`
- `vim`
- `emacs`
- `pico`

Linux - File Streams

- STDOUT
 - Save STDOUT to file
 - `ls > list_of_files.txt`
 - Append STDOUT to file
 - `ls >> list_of_files.txt`
- STDIN
 - Count lines in STDIN
 - `wc -l < list_of_files.txt`
- STDERR
 - Save error messages to file
 - `ls -e 2> errors.txt`

Linux - Piping

Passes output from one command to the input of the next.

`command1 | command2`

- `grep 'pattern' filename | wc -l`

`command1 | command2 | command3 (...etc)`

- `cat filename.csv | cut -f 1 | sort | uniq`

Software Module System

Print available modules

```
module avail
```

Print available modules starting with R

```
module avail R
```

Load default module R

```
module load R
```

Load specific module R version

```
module load R/3.2.0
```

Print list of loaded modules

```
module list
```

Unload module R

```
module unload R
```

Unload specific module R

```
module unload R/3.2.0
```

Environment Variables

- The HPC cluster uses bash as the default shell environment.
- Within this environment, variables can be set and reused.
`MYVAR='Something'`
`export MYVAR='Something'`
`echo $MYVAR`
- Some software utilizes this feature and requires that specific environment variables be set.
 - `$HOME`
 - Contains your home path
 - `$USER`
 - Contains your username
 - `$PATH`
 - Contains paths of executables
 - `$LD_LIBRARY_PATH`
 - Contains paths of dependencies

Environment Variables

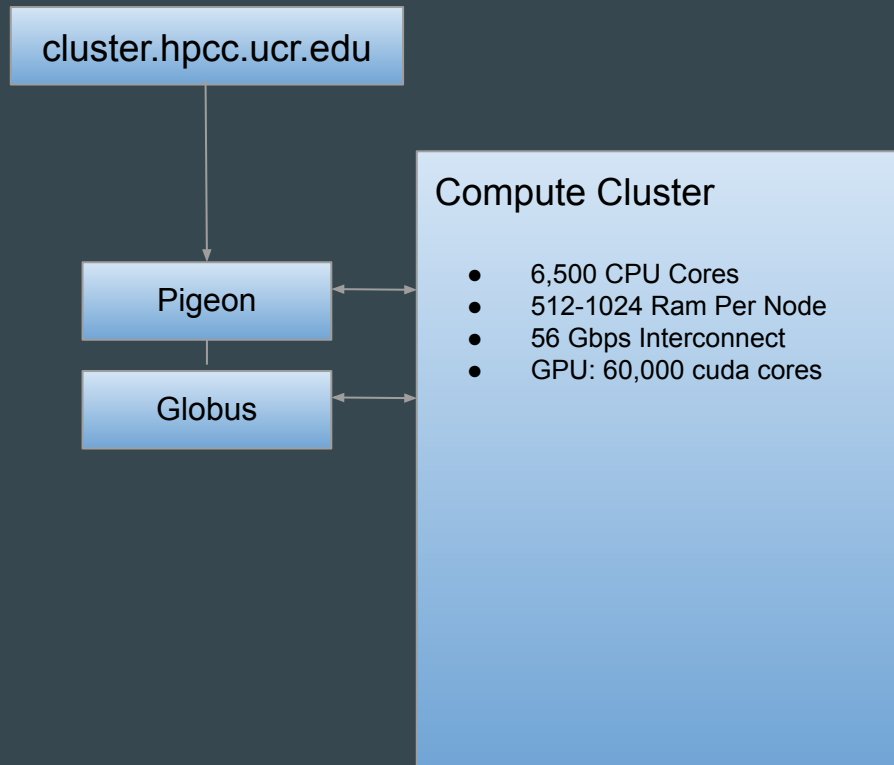
Slurm Job Environment variables (small sample)

- `$_SLURM_SUBMIT_DIR`
 - directory where you ran the script
- `$_SLURM_JOB_ID`
 - the SLURM Job ID of the current Job
- `$_SLURM_NTASKS`
 - number of cpus the job requested
- `$_SLURM_NODELIST`
 - list of nodes this job is running on

Queuing System

Slurm

- Resource Management
- Job Scheduler



Queuing System - Partitions

- Short
 - Nodes: i01-i48
 - Cores: Intel, 256 per user
 - Time (walltime): 2 hours default
 - Time (walltime): 2 hours max
- Intel
 - Default partition
 - Nodes: i01-02, i17-i48
 - Cores: Intel, 256 per user
 - Time (walltime): 168 hours (7 days) default
 - Time (walltime): 30 days max
- Batch
 - Nodes: c01-c48
 - Cores: AMD, 256 per user
 - Time (walltime): 168 hours (7 days) default
 - Time (walltime): 30 days max
- Highmem
 - Nodes: h01-h06
 - Cores: Intel, 32 per user
 - RAM: 100 GB min and 1024 GB max
 - Time (walltime): 48 hours default
 - Time (walltime): 30 days max
- Gpu
 - Nodes: gpu01-gpu05
 - Cores: Intel
 - RAM: 128 GB default
 - Time (walltime): 48 hours default
 - Time (walltime): 30 days max

Queuing System - Partitions

- statsdept
 - Default partition
 - Nodes: i45-i48
 - Cores: Intel, 8 per user
 - Time (walltime): 168 hours (7 days) default
 - Time (walltime): 30 days max

Queuing System - Job Limits

There are both group and user cpu limits for jobs running at the same time.

- Group
 - 512 Cores per group running at any one time
- User
 - 256 Cores per person running at any one time

See your limits:

```
slurm_limits.sh
```

Check total number of cpus currently used by your group in all partitions:

```
group_cpus
```

Queuing System - Status

Current cluster status can be viewed from:

<http://hpcc.ucr.edu/snapshot.html>.

<https://dashboard.hpcc.ucr.edu>

<http://xdmod.hpcc.ucr.edu/>

Command line tool:

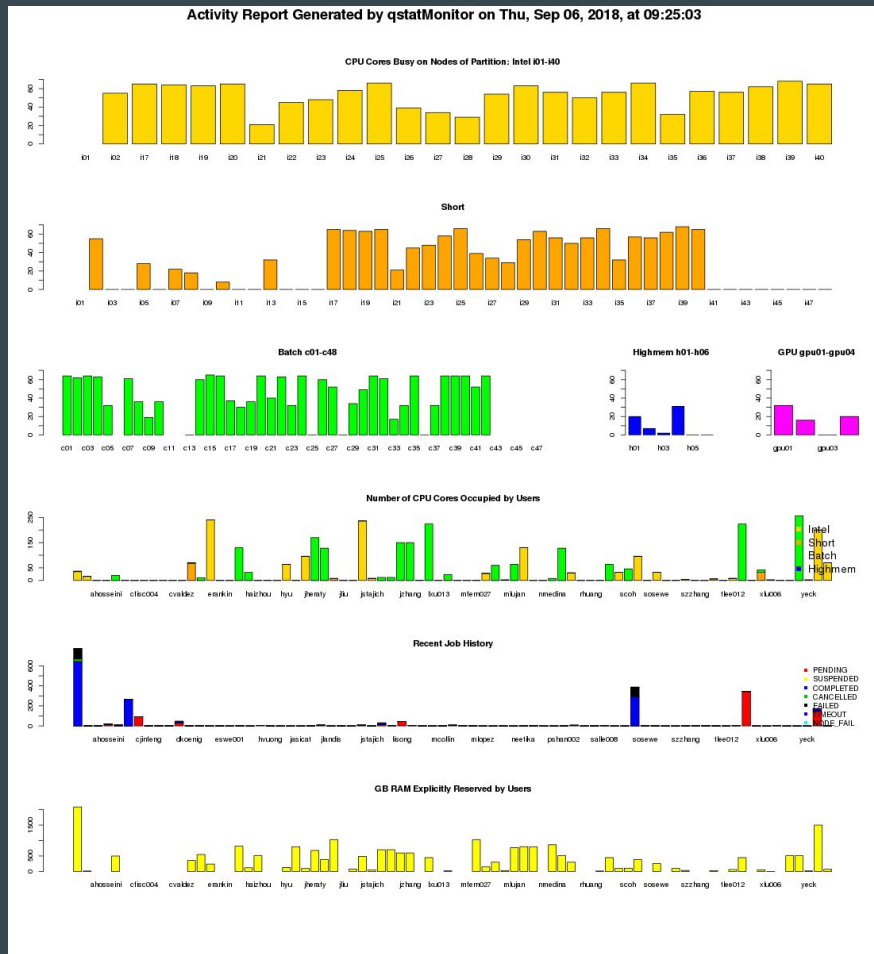
jobMonitor

These commands will also give some interesting information about jobs on the cluster:

sinfo

squeue

sacct



Queuing System - Submitting a Job

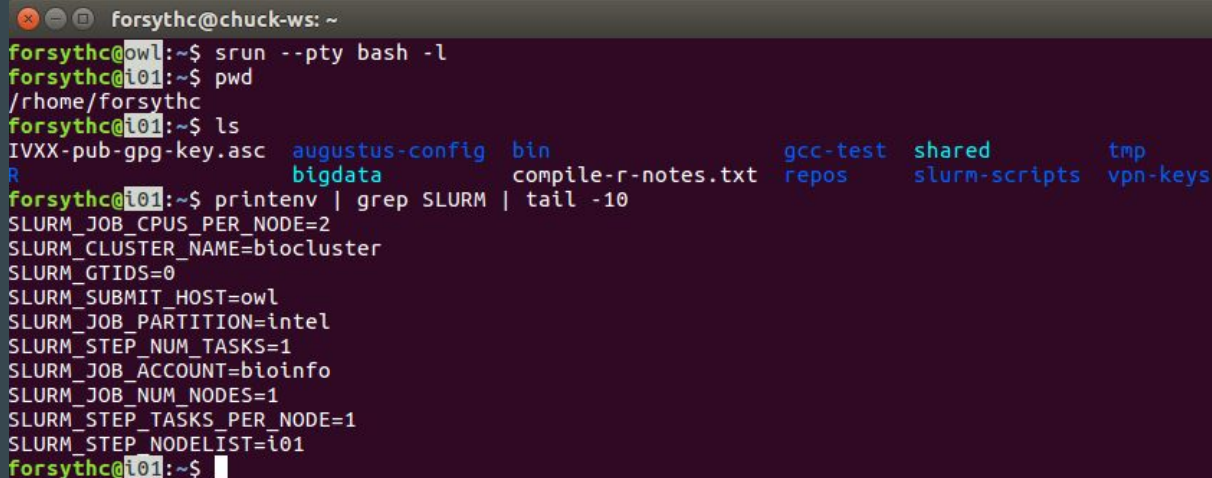
Two major ways to submit jobs on the cluster

- **srun**
 - Used for interactive jobs
 - Examples:
 - `srun --pty bash -l`
 - `srun -p short --x11 --mem=1gb --ntasks 1 --time 02:00:00 --pty bash -l`
- **sbatch**
 - Used to submit batch(non-interactive) jobs
 - Examples:
 - `sbatch sbatch_script.sh`
 - `sbatch --ntasks=12 --mem=24gb sbatch_script.sh`

Queuing System - Submitting a Job

Interactive Job

```
srun -p short --x11 --mem=1gb --ntasks 1 --time 02:00:00 --pty bash -l
```



A terminal window titled 'forsythc@chuck-ws: ~' showing an interactive SLURM job session. The user 'forsythc' is at the 'owl' node. They run 'srun -p short --x11 --mem=1gb --ntasks 1 --time 02:00:00 --pty bash -l' to start an interactive job. The prompt changes to 'forsythc@i01:~\$'. They run 'pwd' and get '/rhome/forsythc'. Then they run 'ls' and see a directory listing: 'IVXX-pub-gpg-key.asc', 'augustus-config', 'bin', 'gcc-test', 'shared', 'tmp', 'bigdata', 'compile-r-notes.txt', 'repos', 'slurm-scripts', and 'vpn-keys'. Next, they run 'printenv | grep SLURM | tail -10' and see SLURM environment variables: 'SLURM_JOB_CPUS_PER_NODE=2', 'SLURM_CLUSTER_NAME=biocluster', 'SLURM_GTIDS=0', 'SLURM_SUBMIT_HOST=owl', 'SLURM_JOB_PARTITION=intel', 'SLURM_STEP_NUM_TASKS=1', 'SLURM_JOB_ACCOUNT=bioinfo', 'SLURM_JOB_NUM_NODES=1', 'SLURM_STEP_TASKS_PER_NODE=1', and 'SLURM_STEP_NODELIST=i01'. The prompt returns to 'forsythc@i01:~\$'.

```
forsythc@chuck-ws: ~  
forsythc@owl:~$ srun -p short --x11 --mem=1gb --ntasks 1 --time 02:00:00 --pty bash -l  
forsythc@i01:~$ pwd  
/rhome/forsythc  
forsythc@i01:~$ ls  
IVXX-pub-gpg-key.asc  augustus-config  bin          gcc-test  shared  tmp  
R                    bigdata         compile-r-notes.txt  repos     slurm-scripts  vpn-keys  
forsythc@i01:~$ printenv | grep SLURM | tail -10  
SLURM_JOB_CPUS_PER_NODE=2  
SLURM_CLUSTER_NAME=biocluster  
SLURM_GTIDS=0  
SLURM_SUBMIT_HOST=owl  
SLURM_JOB_PARTITION=intel  
SLURM_STEP_NUM_TASKS=1  
SLURM_JOB_ACCOUNT=bioinfo  
SLURM_JOB_NUM_NODES=1  
SLURM_STEP_TASKS_PER_NODE=1  
SLURM_STEP_NODELIST=i01  
forsythc@i01:~$
```

Scripting

Basically a series of shell commands:

```
pwd  
mkdir example_dir  
cd example_dir  
touch example_file  
cd ..  
mkdir example_dir2  
cd example_dir2  
touch example_file2  
cd ..  
ls -latr
```

Scripting

Put contents in a file and add the interpreter line to the top - Now it's a Script:

```
#!/bin/bash -l
```

```
pwd
```

```
mkdir example_dir
```

```
cd example_dir
```

```
touch example_file
```

```
cd ..
```

```
mkdir example_dir2
```

```
cd example_dir2
```

```
touch example_file2
```

```
cd ..
```

```
ls -latr
```

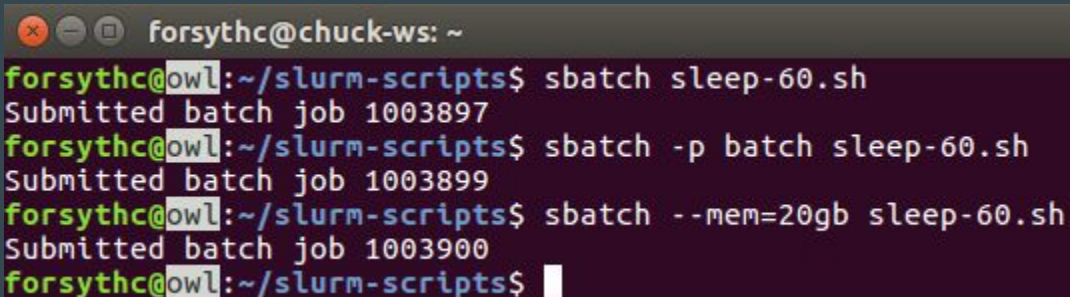

Scripting

- First line in file which defines the interpreter
 - `#!/bin/bash -l`
- Permissions - At least the owner of the file requires execute permissions
 - `chmod u+x myscript.sh`

Queuing System - Submitting a Job

sbatch

- Used to submit batch (non-interactive) jobs
- Examples:
 - `sbatch sbatch_script.sh`
 - `sbatch --ntasks=12 --mem=24gb sbatch_script.sh`



```
forsythc@chuck-ws: ~  
forsythc@owl:~/slurm-scripts$ sbatch sleep-60.sh  
Submitted batch job 1003897  
forsythc@owl:~/slurm-scripts$ sbatch -p batch sleep-60.sh  
Submitted batch job 1003899  
forsythc@owl:~/slurm-scripts$ sbatch --mem=20gb sleep-60.sh  
Submitted batch job 1003900  
forsythc@owl:~/slurm-scripts$
```

Batch Submission Examples

Run the following command to download some example files:

```
git clone https://github.com/ucr-hpcc/hpcc\_intro\_files.git
```

This will create a folder called hpcc_intro_files.

This directory contains example files that you can use to follow along with and run.

Alternate download:

```
wget https://github.com/ucr-hpcc/hpcc\_intro\_files/archive/master.zip
```

Batch Submission Examples

slurm_script_simple.sh

```
#!/bin/bash -l
```

```
#SBATCH --ntasks=1
```

```
#SBATCH --time=00:00:60
```

```
# Print current date
```

```
date
```

```
# sleep for 45 seconds
```

```
/bin/sleep 45
```

```
# Print name of node
```

```
hostname
```

Batch Submission Examples

slurm_script_full_sbatch_options.sh

```
#!/bin/bash -l
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=10 # asking for 10 cpus
```

```
#SBATCH --cpus-per-task=1
```

```
#SBATCH --mem-per-cpu=1G
```

```
#SBATCH --time=0-00:15:00 # 0 day and 15 minutes
```

```
#SBATCH --output=my.stdout
```

```
#SBATCH --mail-user=forsythc@ucr.edu
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --job-name="just_a_test"
```

```
date
```

```
hostname
```

Batch Submission Examples

slurm_script_highmem.sh

```
#!/bin/bash -l
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=2
```

```
#SBATCH --mem=100G
```

```
#SBATCH --time=00-00:00:30
```

```
#SBATCH --job-name="highmem"
```

```
#SBATCH -p highmem
```

```
/bin/sleep 30
```

Batch Submission Examples

slurm_script_gpu.sh

```
#!/bin/bash -l
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=8
```

```
#SBATCH -p gpu
```

```
#SBATCH --gres=gpu:1
```

```
#SBATCH --mem=100G
```

```
#SBATCH --time=00:00:10
```

```
echo "-----"
```

```
hostname
```

```
free -g
```

```
echo "-----"
```

Batch Submission Examples

slurm_script_multinode.sh

```
#!/bin/bash -l

#SBATCH --nodes=2
#SBATCH --ntasks=24
#SBATCH --ntasks-per-node=12
#SBATCH --output=multinode.out
#SBATCH --job-name="multinode test"
#SBATCH --time=00:00:10

echo "-----"
srun hostname
echo "-----"
```


Batch Submission Examples

slurm_script_mpi.sh

```
#!/bin/bash -l

#SBATCH --nodes=4
#SBATCH --ntasks=200
#SBATCH --output=mutinode.out
#SBATCH --job-name="mpi test"
#SBATCH -p batch,short,intel

#SBATCH --time=00:01:00

module load openmpi/2.0.1-slurm-16.05.4
echo "-----"
mpirun mpi_hello_world
echo "-----"
```

Batch Submission Examples

```
slurm_script_pass_argument.sh <input>
```

```
#!/bin/bash -l

#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --job-name="argument example"
#SBATCH --time=00:00:30
#SBATCH -p short,batch,intel

echo $1
```

Queuing System - job output

If no output file is specified in the slurm submission script then the default output file will be produced for each sbatch run.

Format is:

```
slurm-<job id>.out
```

Example:

```
slurm-2378722.out
```

This file contains the STDOUT (standard out) of your submission script.

Queuing System - job info

Check how busy the cluster is overall (shows all jobs in all states):

```
queue -l
```

Check how many jobs overall are in the running state:

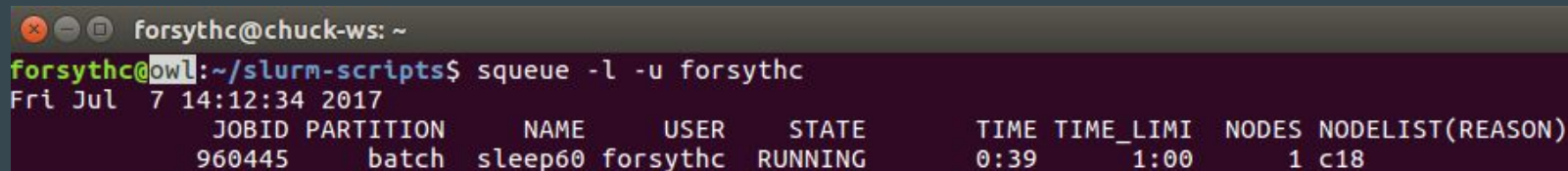
```
queue -l -t R
```

Check how many jobs overall are in the pending state:

```
queue -l -t PD
```

Check the state of your jobs:

```
queue -l -u <user_username>
```



```
forsythc@chuck-ws: ~  
forsythc@owl: ~/slurm-scripts$ queue -l -u forsythc  
Fri Jul 7 14:12:34 2017  
      JOBID PARTITION    NAME    USER    STATE     TIME TIME_LIMI  NODES  NODELIST(REASON)  
      960445    batch  sleep60 forsythc  RUNNING   0:39    1:00      1  c18
```

Queuing System - job info

View recent job info from the accounting database:

```
sacct
```

Lots more this can do:

```
man sacct
```

Queuing System - job control

Cancel your job:

```
scancel <JOBID>
```

Cancel multiple jobs:

```
scancel <JOBID1> <JOBID2> <JOBID3>
```

Cancel ALL your job (caution this will kill all running and queued jobs):

```
scancel -u $USER
```

More info on scancel:

```
man scancel
```

Sharing Files on the Web

Simply move the files into your html directory when you want to share them.

For example, log into the cluster and do the following:

1. Go to your web directory
`cd ~/.html/`
2. Make sure permissions are set correctly
`chmod a+x ~/`
`chmod a+rx ~/.html`
3. Create a default test file
`echo '<h1>Hello!</h1>' > index.html`

Now, test it out by pointing your web-browser to <http://biocluster.ucr.edu/~username/>

Be sure to replace 'username' with your actual user name, and also check permissions on shared directories and parent directories.

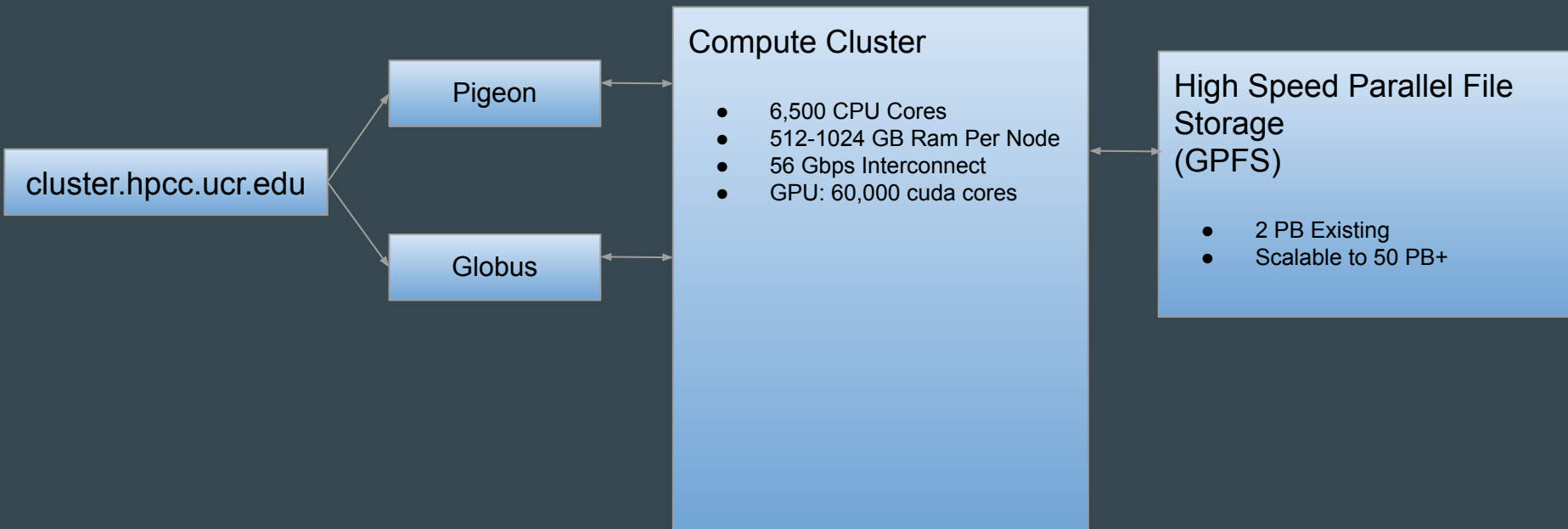
For password protecting html content follow these instructions:

<http://manuals.bioinformatics.ucr.edu/home/hpc#TOC-Password-Protect-Web-Pages>

Helpful Resources

- Command Line
--help, -h, man
- Online Manual
<http://hpcc.ucr.edu>
- Storage Usage
<https://dashboard.hpcc.ucr.edu>
- Announcements
<http://hpcc.ucr.edu/news.html>

Questions



Thank You

Queuing System - Array Jobs

```
#!/bin/bash -l
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=1
```

```
#SBATCH --time=01:00:00
```

```
cd $SLURM_SUBMIT_DIR
```

```
echo "The Slurm Job ID is: "$SLURM_JOBID", The Slurm Array task ID is: "$SLURM_ARRAY_TASK_ID
```

```
inputfilename=inputfile_$(SLURM_ARRAY_TASK_ID).inp
```

```
sbatch --array=1-10 script.slurm  
--array=1855-1900 script.slurm
```

Queuing System - Submitting a Job

Dependency submission (man sbatch)

```
sbatch --dependency=after:JOBID myscript.sh
```

```
sbatch --dependency=afterany:JOBID myscript.sh
```

```
sbatch --dependency=afterok:JOBID myscript.sh
```

```
forsythc@chuck-ws: ~  
forsythc@owl:~/slurm-scripts$ sbatch sleep-60.sh  
Submitted batch job 1003907  
forsythc@owl:~/slurm-scripts$ sbatch --dependency=afterok:1003907 sleep-60.sh  
Submitted batch job 1003908  
forsythc@owl:~/slurm-scripts$ squeue -l -u forsythc  
Mon Jul 17 11:35:54 2017  
      JOBID PARTITION    NAME    USER    STATE       TIME  TIME_LIMI  NODES  NODELIST(REASON)  
      1003907      intel  sleep60  forsythc  PENDING      0:00      1:00      1  (Priority)  
      1003908      intel  sleep60  forsythc  PENDING      0:00      1:00      1  (Dependency)
```

Queuing System - job info detail

Detailed info about a specific job:

scontrol show job <JOBID>

```
forsythc@chuck-ws: ~  
Submitted batch job 969201  
forsythc@pigeon: ~/slurm-scripts$ scontrol show job 969201  
JobId=969201 JobName=sleep60  
  UserId=forsythc(3365) GroupId=bioinfo(1054) MCS_label=N/A  
  Priority=4294207521 Nice=0 Account=bioinfo QOS=normal  
  JobState=RUNNING Reason=None Dependency=(null)  
  Requeue=1 Restarts=0 BatchFlag=1 Reboot=0 ExitCode=0:0  
  RunTime=00:00:06 TimeLimit=00:01:00 TimeMin=N/A  
  SubmitTime=2017-07-10T15:57:02 EligibleTime=2017-07-10T15:57:02  
  StartTime=2017-07-10T15:57:14 EndTime=2017-07-10T15:58:14 Deadline=N/A  
  PreemptTime=None SuspendTime=None SecsPreSuspend=0  
  Partition=intel AllocNode:Sid=pigeon:19625  
  ReqNodeList=(null) ExcNodeList=(null)  
  NodeList=i02  
  BatchHost=i02  
  NumNodes=1 NumCPUs=2 NumTasks=1 CPUs/Task=1 ReqB:S:C:T=0:0:*:*  
  TRES=cpu=2,mem=2G,node=1  
  Socks/Node=* NtasksPerN:B:S:C=0:0:*:* CoreSpec=*  
  MinCPUsNode=1 MinMemoryCPU=1024M MinTmpDiskNode=0  
  Features=(null) Gres=(null) Reservation=(null)  
  OverSubscribe=OK Contiguous=0 Licenses=(null) Network=(null)  
  Command=/rhome/forsythc/slurm-scripts/sleep-60.sh  
  WorkDir=/rhome/forsythc/slurm-scripts  
  StdErr=/rhome/forsythc/slurm-scripts/slurm-969201.out  
  StdIn=/dev/null  
  StdOut=/rhome/forsythc/slurm-scripts/slurm-969201.out  
  Power=
```